

**Part—2**

The Comprehensive LAMP Guide (MySQL)

MySQL is the most widely used Relational Database Management System (RDBMS), and is the 'M' in the LAMP stack. One of the main uses for MySQL is with Web applications. To avoid the database turning into the bottleneck in Web applications, let us set up a MySQL server and tune it for maximum performance.

At the time of writing, the latest (stable) version of MySQL is 5.5.14, and I'll be using that here.

Though MySQL provides binaries for various Linux distributions and even FreeBSD, we're not interested; we need performance, and so we'll be compiling our own. As in the previous article, I do not claim that optimisations will definitely increase performance; sometimes, they can cause degradation as well, because it depends more on hardware, CPU cache and DRAM.

MySQL, unlike the traditional `./configure` build methods, uses CMake, an advanced and easy-to-use build system with a nice UI called `ccmake` (command-line) or `cmake-gui` (the GUI, if you run X). However, we'll specify options at the command line itself. Along with CMake, you'll also need a C++ compiler installed. This usually comes with GCC in most distributions, but sometimes is in a separate package (e.g., Fedora has `gcc` and `gcc-c++`).

To download the MySQL source tarball, navigate to <http://dev.mysql.com/downloads/mysql/5.5.html#downloads> and in the platform drop-down list, choose 'Source Code'. When the download option is displayed, make a note of the

MD5 sum in a text file. After the download is over, to ensure you have the original file and not a tampered version with security holes, verify the file checksum as follows.

Copy the saved MD5 sum to the clipboard, and paste it at the shell prompt, to assign to a variable, as shown; then run the following commands:

```
$ ORG_SUM=<Paste the MD5 sum here>
$ FILE_SUM=$(md5sum mysql-5.5.14.tar.gz | awk -F ' ' '{ print $1 }')
$ [ "$ORG_SUM" == "$FILE_SUM" ] && echo verification success ||
echo verification failed
```

If your download was not okay, re-download it, else read on.

Source code configuration

CMake accepts options on the command line with the `-D` switch; for example, `cmake -D<OPTION NAME>=<OPTION VALUE>`. Significant options while